

One fab, every layer: what IAQ delivers.

A section through IAQ's own Revit coordination model. Every numbered layer is designed, procured, built and proven by the same company.

CSA (Civil, Structural and Architectural)

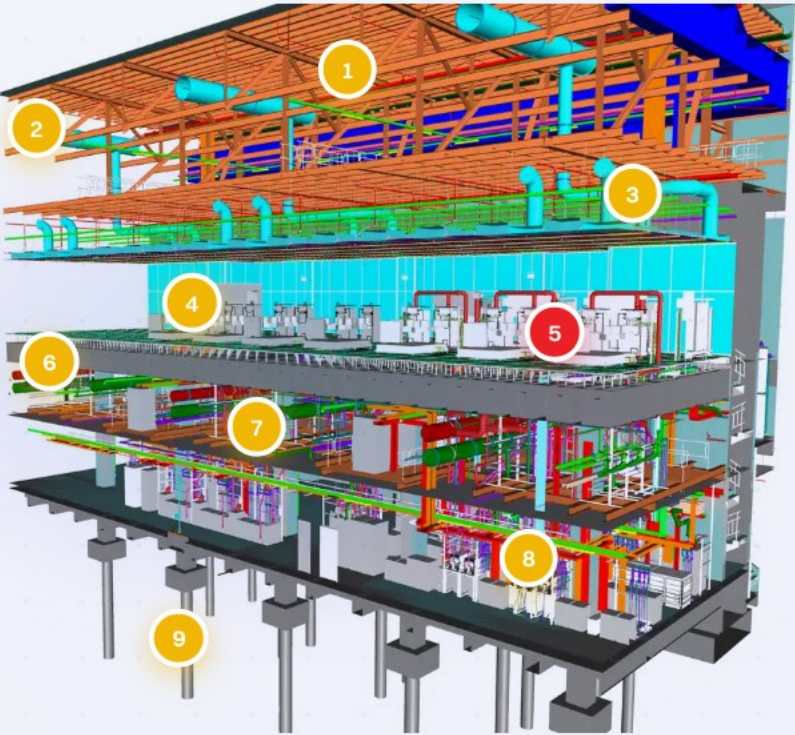
The building

1 Roof steel and structural frame

4 Cleanroom envelope

6 Waffle slab and raised floor

9 Piles and pile caps



MEP (Mechanical, Electrical and Plumbing)

What makes it run

2 Air handling and ducting

3 Building services, the interstitial level

8 Sub-fab plant and fire protection

Process utilities

What the tools run on

7 Process utilities in the sub-fab

Tools hookup

Service 6, the tools themselves

5 Process tools, hooked up

It assembles in the order IAQ builds it

The same model, six frames of sixty one

1 Piles and pile caps

2 Frame, slabs and roof steel

3 Cleanroom envelope

4 Building services

5 Tools and sub-fab plant

6 The whole fab



One fab, every layer

● Business unit ○ Delivery model ● Service ● Work ● System ● Market

Cover

IAQ at a glance: one group, **three business units.**

A total facility solutions provider for controlled environments. Each unit answers one question a facility owner has.



Since 1995

31 years, founded in Malaysia as a cleanroom specialist



450 people

Engineers, project teams and site crews, group wide



250+ projects

Delivered for hi-tech manufacturers and facility owners



1,050,000 m²

Of cleanroom built



7 countries

Eight offices, headquartered in Shah Alam



ISO 3

The cleanest class IAQ has built



Business unit 1

EPC (Engineering, Procurement and Construction)

Builds the facility.

- Design, procurement, construction and commissioning under one contract
- Cleanrooms, dry rooms and hi-tech facilities
- CSA (Civil, Structural and Architectural), MEP (Mechanical, Electrical and Plumbing) and process utilities, in house

EPCC

EPCM



Business unit 2

Process Critical Utilities & Total Tool Installation Solutions

Re-equips a live semiconductor fab.

- Utilities: gas, chemical and slurry, water, process exhaust
- Tool installation: facilitate, rig in, hook up, commission
- Semiconductor fabs, front end to back end

Standalone

EPC or EPCM model



Business unit 3

EFM

Runs and maintains it.

- Energy Facility Management, a registered ESCO (Energy Service Company)
- IAQ funds the upgrade, paid from the savings
- Chiller plant and district cooling, run under a service level

Cooling as a Service

Energy Performance Contracting

Seven markets

 Semiconductor

 Data Centre

 EV Battery

 Photovoltaics

 District Cooling & Heating

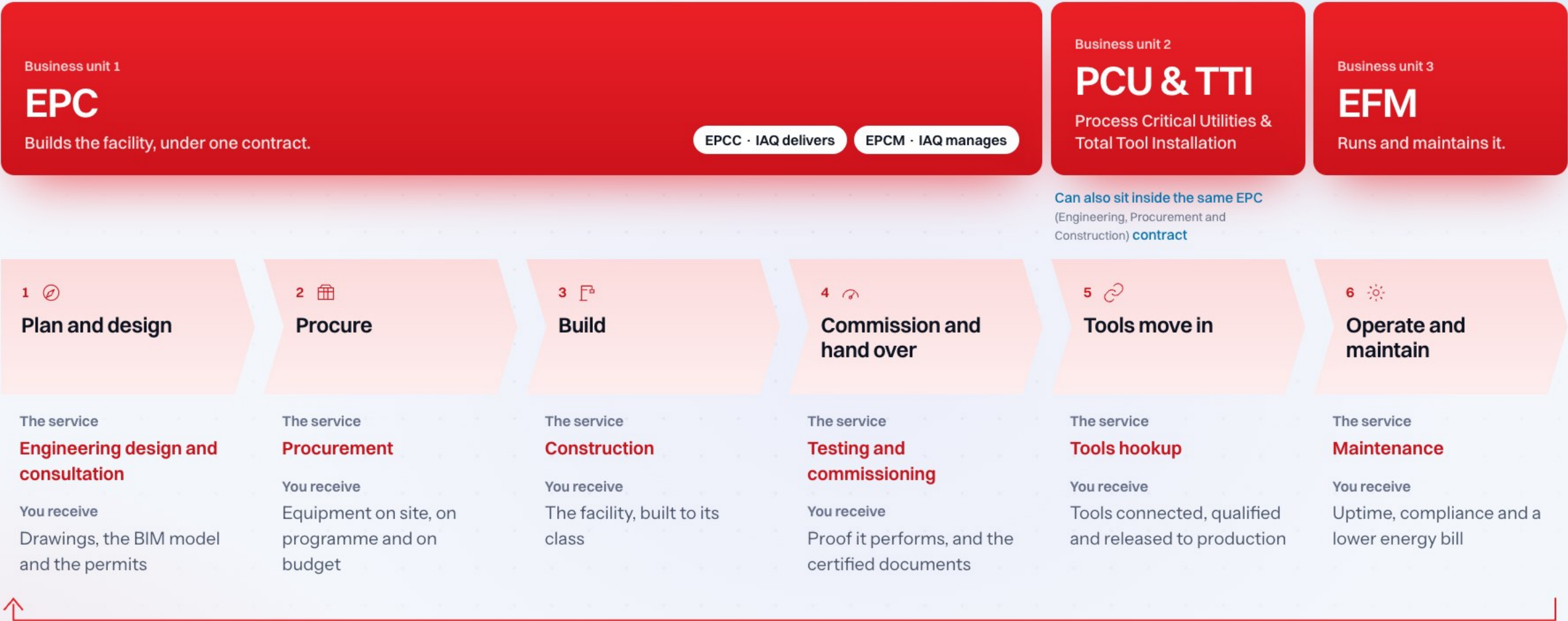
 Bio LifeScience

 Food & Beverage

Short forms: EPCC (Engineering, Procurement, Construction and Commissioning) · EPCM (Engineering, Procurement and Construction Management)

Three units, across the life of **your facility**.

Read it as the owner does: left to right, from the first drawing to a plant in operation, and round again at the next expansion.

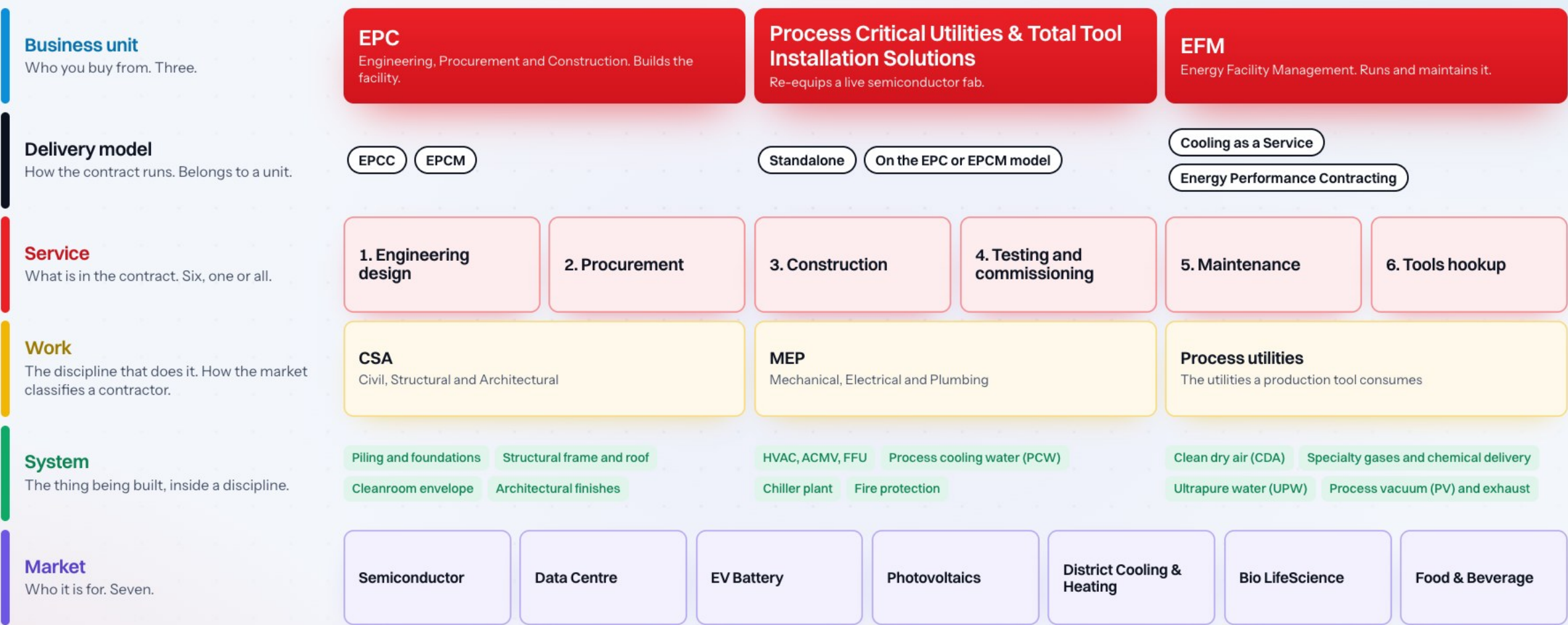


And round again. New tools, a new line or an energy upgrade start the next design. The same three units carry it.

Short forms: EPCC (Engineering, Procurement, Construction and Commissioning) · EPCM (Engineering, Procurement and Construction Management) · EFM (Energy Facility Management) · BIM (Building Information Modelling)

IAQ in one picture: which kind of thing is which.

Six kinds of thing, one colour each, the same on every slide. Read down: who you buy from, how the contract runs, what is in it, who does it, what gets built, and for whom.



Short forms: EPCC (Engineering, Procurement, Construction and Commissioning) · EPCM (Engineering, Procurement and Construction Management) · HVAC (heating, ventilation and air conditioning) · ACMV (air conditioning and mechanical ventilation) · FFU (fan filter unit)

Three words that mean two things.

Found by cross-checking IAQ's own documents against the 17 September review and the site. The colour says which kind of thing each meaning is.

EPC

One word, used for three different kinds of thing. This is where most of the confusion starts.

Business unit	EPC (Engineering, Procurement and Construction), unit 1 The unit that builds the facility.
Delivery model	The EPC model: EPCC (Engineering, Procurement, Construction and Commissioning), EPCM (Engineering, Procurement and Construction Management) How a contract runs. The hookup unit works on it too.
A contract	Energy Performance Contracting EFM (Energy Facility Management)'s contract. Always written in full, never EPC.

Say it this way

Say “the EPC unit” or “the EPC model”. Never shorten the energy contract.

Hookup

Unit 2 went by two names, and one of them borrowed the name of a service.

Business unit	Process Critical Utilities & Total Tool Installation Solutions Unit 2, IAQ's own written name. “Total Tools Hookup Solution” is retired.
Service	Tools hookup, service 6 The work of connecting a tool. The only thing “hookup” names now.

Say it this way

Name the unit by what it sells. Keep “hookup” for the work.

PCW, CDA, UPW

Systems turn up in two places, so they get listed as services by mistake.

System	Built with the facility PCW (process cooling water) is mechanical work. CDA (clean dry air) and UPW (ultrapure water) are process work.
Service	Their last metres are hookup From the main to the tool, service 6 connects them.

Say it this way

A system is never a service. It appears twice because two crews touch it.

Six services. Buy one, or all of them.

IAQ calls the six its total solution: the scope written into the contract, in the order a project runs. Under it, the business unit that carries each one.



Every discipline

Civil, structural and architectural, MEP (Mechanical, Electrical and Plumbing), and process utilities, in house.



Any combination

One service, several, or all six. The model that governs it stays the same.



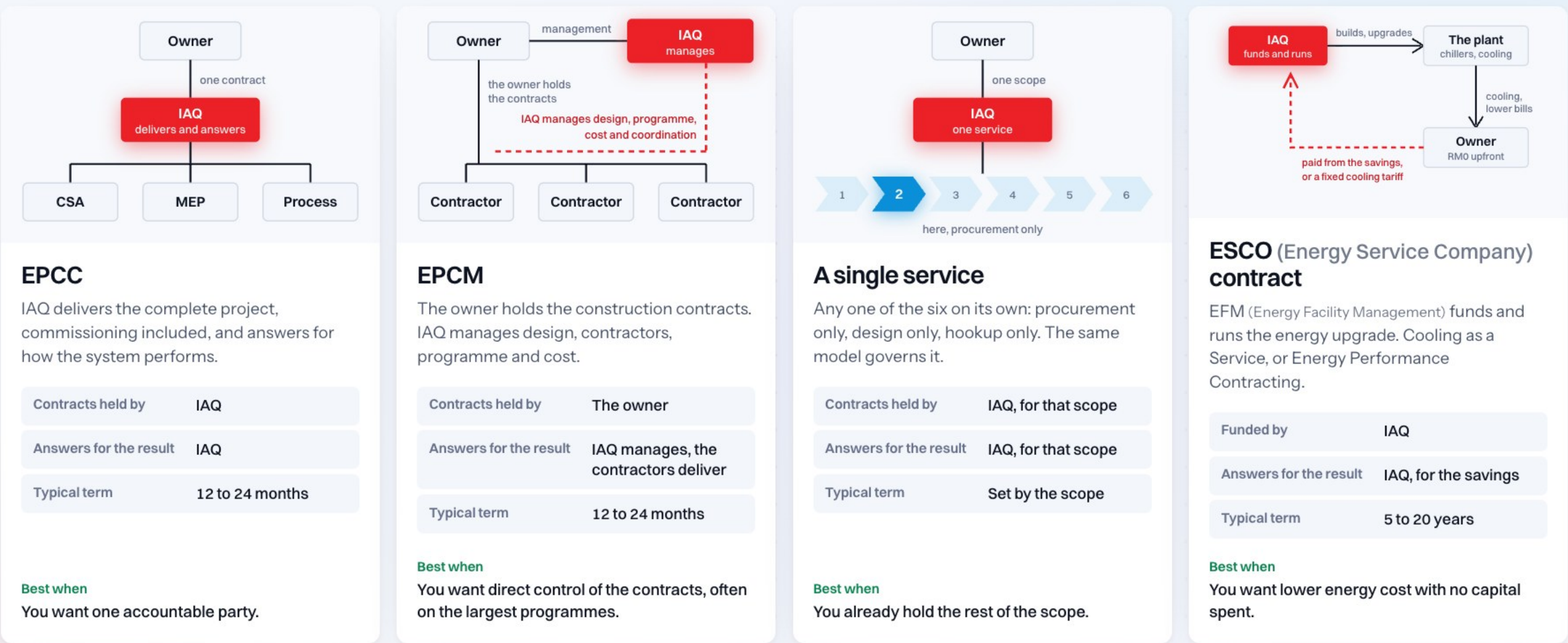
Two delivery models

EPCC, IAQ delivers the whole project. EPCM, IAQ manages it for the owner.

Short forms: BIM (Building Information Modelling) · EPC (Engineering, Procurement and Construction) · PCU & TTI (Process Critical Utilities & Total Tool Installation) · EFM (Energy Facility Management) · EPCC (Engineering, Procurement, Construction and Commissioning) · EPCM (Engineering, Procurement and Construction Management)

Four ways to contract IAQ.

The model governs the work. The scope, one service or all six, sits inside it. In EPCC (Engineering, Procurement, Construction and Commissioning), IAQ does the work. In EPCM (Engineering, Procurement and Construction Management), IAQ manages the work for the owner.



Short forms: CSA (Civil, Structural and Architectural) · MEP (Mechanical, Electrical and Plumbing)

Three disciplines. One contractor.

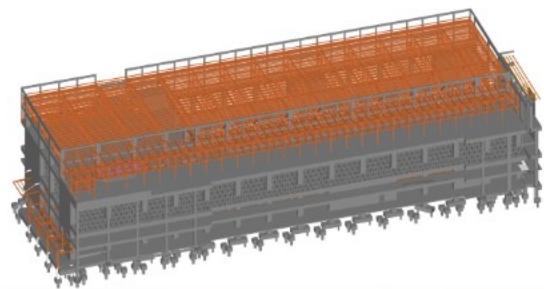
The market classifies a contractor by the work it delivers. A specialist carries one discipline. IAQ carries all three. The drawings are IAQ's own Revit model.

CSA

Civil, Structural and Architectural

The building itself.

- Piling and foundations
- Structural frame and roof
- Cleanroom envelope: walls, ceiling grid, raised floor
- Architectural finishes

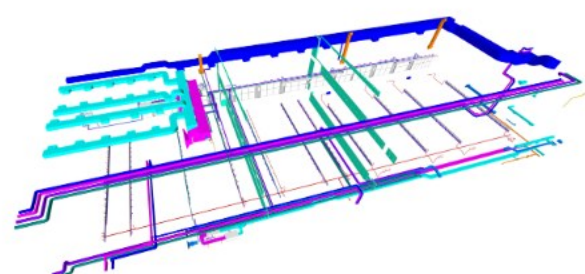


MEP

Mechanical, Electrical and Plumbing

What makes the building run.

- HVAC, ACMV and fan filter units
- Process cooling water (PCW)
- Chiller plant and district cooling
- Fire protection
- Electrical, high and low voltage
- Plumbing and drainage

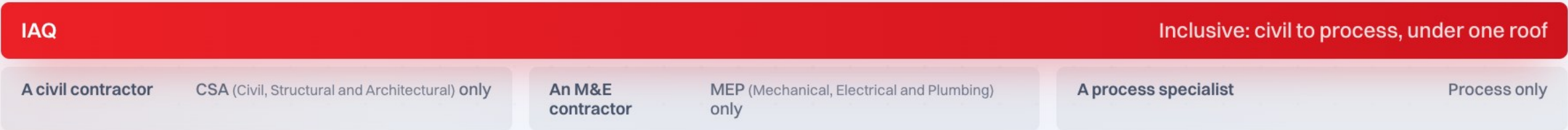
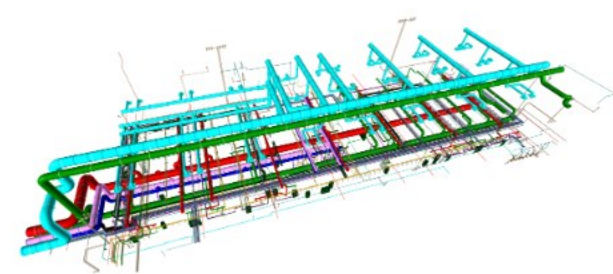


Process utilities

The utilities a production tool consumes

What the tool runs on.

- Clean dry air (CDA)
- Specialty gases and chemical delivery
- Ultrapure water (UPW)
- Process vacuum (PV) and exhaust
- Waste treatment



Short forms: HVAC (heating, ventilation and air conditioning) · ACMV (air conditioning and mechanical ventilation) · M&E (mechanical and electrical)

EPC: one contract, from first drawing to final handover.

Engineering, Procurement and Construction. IAQ designs the system, procures it, builds it and proves it operates as intended.

1

point of contact for the whole facility

12 to 24

months, a typical EPC programme

6

services, from design to tools hookup

3

disciplines in house: CSA, MEP, process

The diagram shows a hierarchical structure. At the top is a box labeled 'Owner'. A vertical line connects 'Owner' to a red box labeled 'IAQ delivers and answers'. This red box is labeled 'one contract'. Below the red box, a horizontal line branches into three vertical lines, each connecting to a box labeled 'CSA', 'MEP', and 'Process' respectively.

EPCC (Engineering, Procurement, Construction and Commissioning)

Turnkey. IAQ delivers the complete project and is accountable for how the system performs.

The diagram shows a hierarchical structure. At the top is a box labeled 'Owner'. A horizontal line connects 'Owner' to a red box labeled 'IAQ manages', with the label 'management' above the line. Below 'Owner', a vertical line connects to a horizontal line that branches into three vertical lines, each connecting to a box labeled 'Contractor'. A dashed red line connects the 'IAQ manages' box to the horizontal line above the 'Contractor' boxes. Text labels include 'the owner holds the contracts' near the Owner-Contractor line, and 'IAQ manages design, programme, cost and coordination' near the dashed line.

EPCM (Engineering, Procurement and Construction Management)

Managed for the owner, who keeps the construction contracts and direct control.

At a glance	EPCC	EPCM
Who holds the construction contracts	IAQ	The owner
Who answers for the system performing	IAQ	IAQ manages, the owner's contractors deliver
Commissioning and start-up	Included	Coordinated by IAQ
Typical programme	12 to 24 months	12 to 24 months

Short forms: CSA (Civil, Structural and Architectural) · MEP (Mechanical, Electrical and Plumbing)

From a delivered tool to a producing tool.

Two halves, one unit, for semiconductor fabs only. Process Critical Utilities builds the utilities with the base build. Total Tool Installation then connects each tool to them.

Process Critical Utilities

IAQ's five solution groups

Water

UPW (ultrapure water) plant and distribution, process drains, waste water treatment, reclaim

Chemical and slurry

Bulk chemical, blending, dispense, storage tanks

Gas

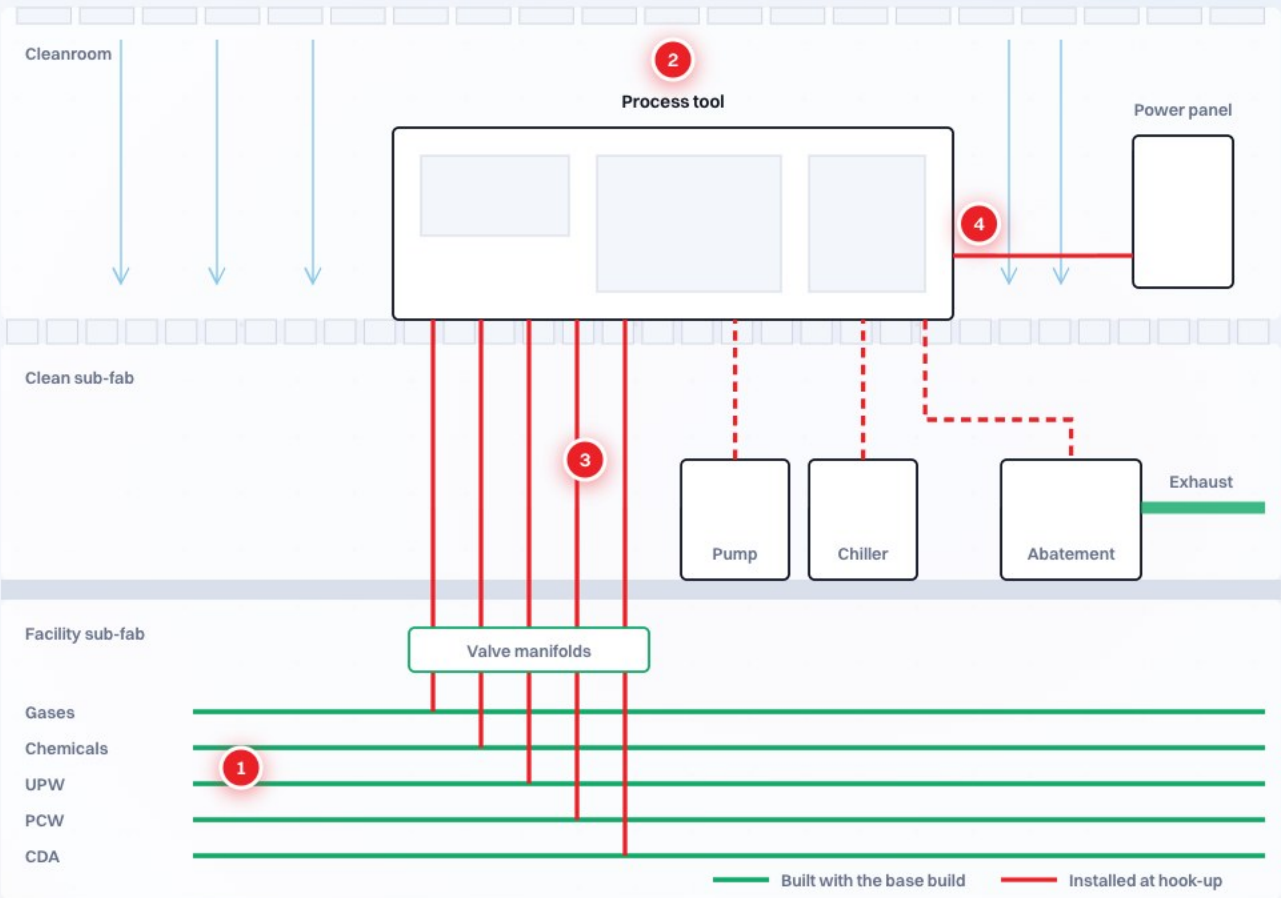
Bulk and specialty gas, abatement, bunkers, liquid dispense

Process exhaust

General, acid, caustic, solvent and calamity

Space management

The sub-fab planned in 3D BIM (Building Information Modelling), off-site modules



Total Tool Installation

IAQ's four phases, marked on the section

- 1 Facilitization**
Before the tool arrives: drops, power, structure and room built to the maker's requirements.
- 2 Rig-in and move-in**
Uncrated in an airlock, moved on an agreed route, set, levelled and isolated.
- 3 Hook-up**
Each utility tapped from its valve manifold and connected into the tool.
- 4 Commissioning**
Leak tested, purged, balanced, trip tested, then handed to operations.

How it is bought

On its own, even in a fab IAQ did not build, or inside an EPCC (Engineering, Procurement, Construction and Commissioning) or EPCM (Engineering, Procurement and Construction Management) contract.

Short forms: PCW (process cooling water) · CDA (clean dry air)

EFM: lower energy bills, funded by IAQ.

Energy Facility Management, a registered ESCO (Energy Service Company) under Suruhanjaya Tenaga. IAQ funds, builds and runs the upgrade.

Two ways to pay for it

10 to 20 years

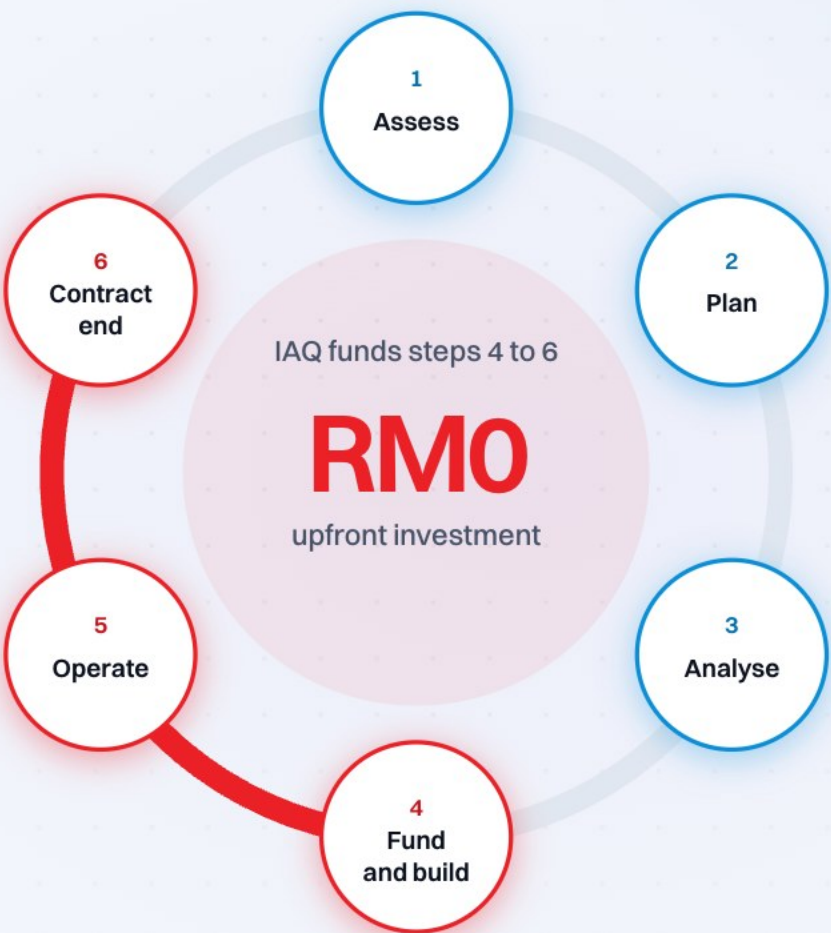
Cooling as a Service

IAQ finances, builds or rehabilitates, and operates the chiller plant. A fixed tariff for the cooling used, under a service level agreement. The plant is handed over at the end.

5 to 10 years

Energy Performance Contracting

IAQ funds and delivers the upgrades. Every payment comes from the savings actually measured. After the contract, the owner keeps all of them.



The four problems it solves

High electricity bills

Inefficient cooling, driving cost up year after year.

Rising carbon emissions

Compliance risk against regulation and targets.

Ageing infrastructure

Chiller plant past its best, and a capital bill to replace it.

Operational disruption

Downtime from old systems and poor maintenance.

Proof, by market and by the work delivered.

The 18 published references of 250+ delivered projects. Each light is one project; client names are withheld.



The full record is 250+ projects. This slide grows with the approved project list: every project added to the registry lights here on its own.